Exercise - 1.4

1. For the curve $y = x^3$ given in Figure 1.67, draw

   (i) $y = -x^3$ (ii) $y = x^3 + 1$ (iii) $y = x^3 - 1$ (iv) $y = (x + 1)^3$ with the same scale.

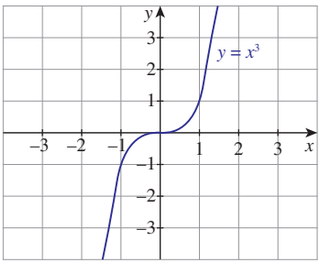


Figure 1.67

2. For the curve $y = x^{(\frac{1}{3})}$ given in Figure 1.68, draw

   (i) $y = -x^{(\frac{1}{3})}$ (ii) $y = x^{(\frac{1}{3})} + 1$ (iii) $y = x^{(\frac{1}{3})} - 1$ (iv) $y = (x + 1)^{(\frac{1}{3})}$

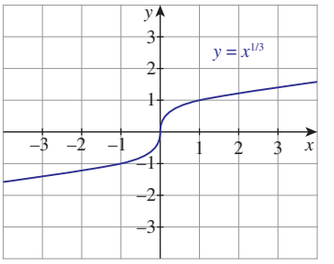


Figure 1.68

3. Graph the functions $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$ on the same coordinate plane. Find $f \circ g$ and graph it on the plane as well. Explain your results.
4. Write the steps to obtain the graph of the function $y = 3(x - 1)^2 + 5$ from the graph $y = x^2$.
5. From the curve $y = \sin x$, graph the functions

   (i) $y = \sin(-x)$ (ii) $y = -\sin(-x)$ (iii) $y = \sin\left(\frac{\pi}{2} + x\right)$ which is $\cos x$

   (iv) $y = \sin\left(\frac{\pi}{2} - x\right)$ which is also $\cos x$ (refer trigonometry)
6. From the curve $y = x$, draw

   (i) $y = -x$ (ii) $y = 2x$ (iii) $y = x + 1$ (iv) $y = \frac{1}{2}x + 1$ (v) $2x + y + 3 = 0$.
7. From the curve $y = |x|$, draw (i) $y = |x - 1| + 1$ (ii) $y = |x + 1| - 1$ (iii) $y = |x + 2| - 3$.
8. From the curve $y = \sin x$, draw $y = \sin|x|$ (Hint: $\sin(-x) = -\sin x$.)